



Fundamentals of Software Engineering

LECTURE 7: USER STORIES

USER STORIES AND SPLITTING USER STORIES INTO SUB-USER STORIES

User Stories

seek to combine the strengths
of written and verbal communication,
where possible supported by a picture.

** Kent Beck coined the term user stories in Extreme
Programming Explained 1st Edition, 1999*

What is a User Story?

A concise, written description of a piece of functionality that will be valuable to a user (or owner) of the software.

Stories are:

- User's needs
- Product descriptions
- Planning items
- Tokens for a conversation
- Mechanisms for deferring conversation

User Story is written for whom?

User story is written from the perspective of stakeholder and **NOT DEVELOPER OR TESTER**

So it has to be

- Simple and not technical (task is technical)
- Understandable
- Written in natural language

From the perspective of

- client/customer/user
- Admin

User Story (Cards) have 3 parts

1. **Description** - A written **description** of the user story for planning purposes and as a reminder
2. **Conversation** - A section for capturing **further information** about the user story and **details** of any conversations
3. **Confirmation** - A section to convey what **tests** will be carried out to confirm the user story is complete and working as expected

User Story Template – 1

- As a [user role] I want to [goal] so I can [reason]
- As a [type of user] I want to [perform some task] so that I can [reach some goal]

Example:

- As a registered user I want to log in so I can access subscriber-only content

User Story Template – 2

Who (user role)

- **What** (goal)

Why (reason)

- gives clarity as to why a feature is useful
- can influence how a feature should function
- can give you ideas for other useful features that support the user's goals

User Story Description

Steps:

- Start with a **title**.
- Add a **concise description** using the templates.
- Add other relevant **notes, specifications, or sketches**
- Before building software write **acceptance criteria** (*how do we know when we're done?*)

Example: Front of Card

#0001	USER LOGIN	Fibonacci Size # 3
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As a [registered user], I want to [log in], so I can [access subscriber content].

For new features, annotated wireframe. For bugs, steps to reproduce with screenshot. For non-functional stories, explain scope/standards.

User Login

Username:

Password:

Remember me ☐

[message]

[Forgot password?](#)

Store cookie if ticked and login successful.

User's email address. Validate format.

Authenticate against SRS using new web service.

Go to forgotten password page.

Display message here if not successful. (see confirmation scenarios over)

Further information is attached to this story on VSTS Product Backlog.

Example: Back of Card

Confirmation

1. Success – valid user logged in and referred to home page.
 - a. 'Remember me' ticked – store cookie / automatic login next time.
 - b. 'Remember me' not ticked – force login next time.
2. Failure – display message:
 - a) "Email address in wrong format"
 - b) "Unrecognised user name, please try again"
 - c) "Incorrect password, please try again"
 - d) "Service unavailable, please try again"
 - e) Account has expired – refer to account renewal sales page.

How detailed should a User Story be?

Detailed enough

For the team to start working from

To establish further details and clarifications
at the time of development.

How detailed should a User Story be?

1. As customer I can log-in with my account, so I don't have to re-enter my personal — information every time;
2. As customer I can review and confirm my order, so I can correct mistakes before I pay;
3. As customer I can pay for my order with a wire transfer, so that I can confirm my order;
4. As customer I can pay for my order with credit card, so that I can confirm my order;
5. As customer I receive a confirmation email with my order, so I have proof of my purchase; (acceptance criteria)

How is detail added to user stories?

Detail can be added to user stories in two ways:

- By splitting a user story into multiple, smaller user stories.
- By adding “conditions of satisfaction.”

Ex1: As a vice president of marketing, I want to select a holiday season to be used when reviewing the performance of past advertising campaigns so that I can identify profitable ones.

Detail could be added as:

- Make sure it works with major retail holidays: Christmas, Easter, President’s Day, Mother’s Day, Father’s Day.
- Holiday seasons can be set from one holiday to the next (such as Thanksgiving to Christmas).
- Holiday seasons can be set to be a number of days prior to the holiday.

Splitting user story into sub-stories

1. As a vice president of marketing, I want to select major retail holidays so I can review campaign performance for those holidays.
2. As a vice president of marketing, I want to select a holiday season from one holiday to another (e.g., Thanksgiving to Christmas) so I can analyze campaigns in that period.
3. As a vice president of marketing, I want to define a holiday season as a number of days before a holiday so I can see early campaign results.

Accepting criteria for user story 1

Acceptance Criteria (Back of User Story Card)

1. Holiday Selection

The system provides a predefined list of major retail holidays (e.g., Black Friday, Christmas, Easter, etc.).

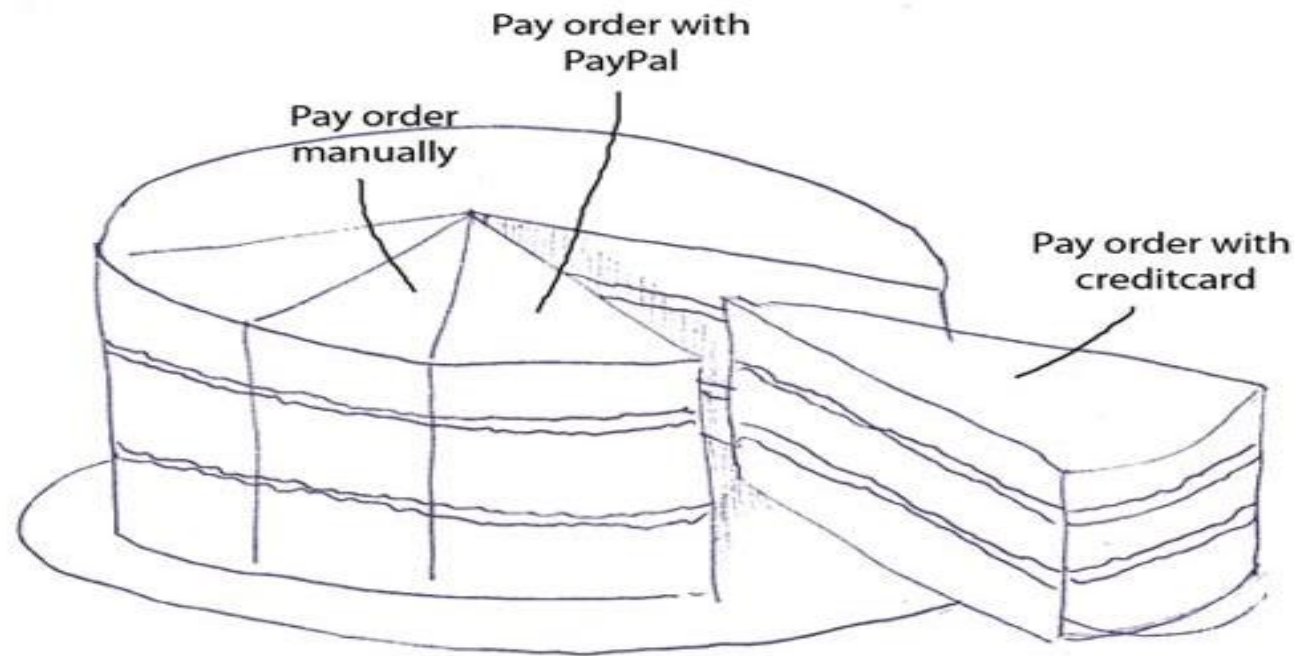
The Vice President of Marketing can select one or more holidays from the list.

Only valid, predefined holidays can be selected.

	Story ID:		Story Title:
User Story:		Importance:	
As a: <role> I want: <some goal> So that: <some reason>			
Acceptance Criteria		Estimate:	
And I know I am done when:			
		Type:	
		☐ Search	
		☐ Workflow	
		☐ Manage Data	
		☐ Payment	
		☐ Report/ View	

How to split a User Story into multiple user stories ?

A friendlier approach is to slice the cake up in vertical slices, each with a bit of cake, cream and fruit



Splitting User story w.r.t business rules an example

1. As shop owner I can decline orders below 10 dollars, because they aren't profitable;
2. As shop owner I can decline customers from outside the US, because the shipping expenses make these orders unprofitable;
3. As shop owner I can reserve ordered products from stock for 48 hours, so other customers see a realistic stock;
4. As shop owner I can automatically cancel orders for which I have not received payment within 48 hours, so I can sell them again to other customers;

INVEST in Good User Stories

Independent – User Stories should be as independent as possible.

Negotiable – a User Story is not a contract. It is not a detailed specification. It is a reminder of features for the team to discuss and collaborate to clarify the details near the time of development.

Valuable – User Stories should be valuable to the user (or owner) of the solution. They should be written in user language. They should be features, not tasks.

Estimatable – User Stories need to be possible to estimate. They need to provide enough information to estimate, without being too detailed.

Small – User Stories should be small. Not too small and not too big.

Testable – User Stories need to be worded in a way that is testable, i.e. not too subjective and to provide clear details of how the User Story will be tested.

User Stories Summary

User Stories combine written and verbal communications, supported with a picture where possible.

User Stories should describe features that are of value to the user, written in the user's language.

User Stories detail just enough information and no more.

Details are deferred and captured through collaboration just in time for development.

Test cases should be written before development, when the User Story is written.

User Stories should be Independent, Negotiable, Valuable, Estimatable, Small and Testable.



Thank
You

A blue paper cutout of the words "Thank You" in a stylized, rounded font. The text is white with a blue outline. The cutout is hanging from a thin brown string that is tied in a loop at the top. The background is white.